

DIESTEL'S LWCG VERSUS WCG QUESTION FROM ARXIV:1512.08628 IS ANSWERED BY ARXIV:1706.08147

FA.BANACH.001, AGENT_SUPER.01

STATUS

This is a literature-already-answered packet. Aviles–Guirao–Lajara–Rodriguez–Tradacete [1] state Diestel's question: is every lattice weakly compactly generated Banach lattice weakly compactly generated as a Banach space? A later paper of Aviles–Rodriguez–Tradacete [2] explicitly gives a negative answer.

SOURCE QUESTION

In Section 1 of [1], the authors define a Banach lattice X to be lattice weakly compactly generated (LWCG) if $X = L(K)$ for some weakly compact $K \subset X$. They record the implications

$$\text{WCG} \Rightarrow \text{LWCG} \Rightarrow \text{IWCG} \Rightarrow \text{BWCG},$$

and state that the equivalence between LWCG and WCG for general Banach lattices was an open question raised by J. Diestel at the 2011 La Manga conference.

SUPPORTING ANSWER

The introduction of [2] restates the problem as Problem 1.1:

Is every LWCG Banach lattice WCG?

It then says that the paper provides a negative answer. The precise result is Corollary 5.5 of [2]: if Γ is uncountable and $1 < p \leq 2$, then

$$\text{FBL}[\ell_p(\Gamma)]$$

is LWCG but is not isomorphic to a subspace of any WCG Banach space. In particular it is not WCG. Taking $p = 2$ gives a direct counterexample to Diestel's question as quoted in [1].

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SCOPE

This answers the general LWCG = WCG question negatively. It does not make the other structural questions in [1] obsolete, such as the separate question of whether an LWCG Banach lattice can contain a sublattice isomorphic to ℓ_∞ , or the positive order-continuous/Dedekind complete subcases proved in the source.

SEARCH NOTES

The cheap run indexes had no packet for arXiv:1512.08628 itself. A nearby packet records that the residual final questions of [2] about $\text{FBL}[c_0(\Gamma)]$ and $\text{FBL}[\ell_p(\Gamma)]$, $p > 2$, are answered by arXiv:1806.02553. That is distinct from this packet, which records the earlier negative answer to Diestel's original LWCG versus WCG question from [1].

REFERENCES

- [1] A. Aviles, A. J. Guirao, S. Lajara, J. Rodriguez, and P. Tradacete, *Weakly compactly generated Banach lattices*, Studia Math. 234 (2016), no. 2, 165–183; arXiv:1512.08628.
- [2] A. Aviles, J. Rodriguez, and P. Tradacete, *The free Banach lattice generated by a Banach space*, J. Funct. Anal. 274 (2018), no. 10, 2955–2977; arXiv:1706.08147.